INCH-POUND

MIL-I-27619G
AMENDMENT 3

22 February 1990
SUPERSEDING
AMENDMENT 2
17 FEBRUARY 1984

MILITARY SPECIFICATION

INDICATORS, ATTITUDE DIRECTOR

This amendment forms a part of MIL-I-27619G, dated 2 August 1974, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1 Delete "first sentence" and add: "Government documents."

* Add: "2.1.1 Specifications and standards. The following specifications and standards, form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specificatons and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)."

Under SPECIFICATIONS, Federal, delete "QQ-P-416".

Under SPECIFICATIONS, Military, delete "MIL-I-6181" and add: "MIL-M-7793/6 Meter, Time, Totalizing, Miniature, Digital, 115 Volt, 400 Hertz".

Add the following boxed paragraph to bottom of page:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEOR, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

Under SPECIFICATION, Military, delete "MIL-B-83292", and add: "MIL-B-81793 Bearing Ball, Precision, For Instruments And Rotating Components" and "MIL-C-83488 Coating, Aluminum, Ion Vapor Deposited".

* Under STANDARDS, Military, delete "MIL-STD-143", "MIL-STD-794", "MIL-STD-831", MS17322" and add: MIL-STD-461 Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference", "MIL-STD-462 Electromagnetic Interference Characteristics, Measurement Of", "MIL-STD-970 Standards And Specifications, Order Of Preference For The Selection Of",

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"MIL-STD-2073-1A DOD Material Procedures For Development And Applicaton Of Packaging Requirements" and change "MIL-STD-810" to MIL-STD-810C".

* Delete the parenthetical source statement that follows the listing of Government Specification and Standards and add: "(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099)"

* Add: (Copies of MIL-STD-810C is available from Director, Naval Publications and Printing Service Office, 700 Robins Ave, Bldg 4 Section D, Philadelphia PA 19111).

* Add:"2.2 Non-Government publications. The following document (s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2)."

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)"

* Add "2.3 Order of Precedence. In the event of a conflict between the text of this document and references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

PAGE 3

* 3.2 Delete entirely.

* 3.3 Line 3 delete "MIL-STD-143" and add: "MIL-STD-970".

PAGE 4

* Add new "3.4.6 Recycled and reclaimed materials. Recycled and reclaimed materials shall be used to the maximum extent possible without jeopardizing the intended end use of the item."

PAGE 5

3.5.1.3 Delete entirely and add: "3.5.1.3 Reliability. The indicator shall have a lower test mean-time-between-failure (MTBF) of 1,000 hours at a 90 percent confidence level and a discrimination ratio of 1.5:1."

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PAGE 6

* 3.6.1 Line 2 delete "MIL-I-6181" and add: "MIL-STD-461 and MIL-STD-462".

3.6.5 Delete entirely and add: "3.6.5 Insulation resistance and dielectric. The indicator shall withstand a potential of 500 VDC applied between isolated pins and between pins of the connector and the indicator case for ten seconds. The resistance shall be 10 megohms or greater. The indicator shall also withstand a potential of 200 VAC at a commercial frequency (60 Hz) applied between isolated pins of the connector and between pins of the connector and indicator case for a period of one minute without arc over, breakdown, or leakage current in excess of 0.05 ma."

PAGE 15

3.10.7.2 Delete entirely and add: "3.10.7.2 Case isolation. The case shall be DC and AC isolated from all electrical circuits therein."

PAGE 20

* 3.10.11 Line 2 delete "MS17322" and add: "MIL-M-7793/6".

PAGE 22

* 3.15.1 Line 2 delete "MIL-B-83292" and add: "MIL-B-81793".

* 3.18.2 Delete entirely and add new: "3.18.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, in accordance with MIL-C-83488, Type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the cognizant engineering activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

PAGE 23

* 4.1 Delete paragraph entirely and add new: 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* Add: "4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specificaton shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specificaton shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does

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not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material."

PAGE 24

* 4.2,a,b Delete entirely and add new: "4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4)
b. Quality conformance inspection (see 4.5)."

* Delete entirety and add new "4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.3.1 of this specification."

* 4.4 Delete entirety and add new "4.4 Qualification inspection".

PAGE 25

* 4.4.1.1 Delete entirely.

* 4.4.2, a, b and c delete entirely.

* 4.5 Delete "tests", in "title and line 1" and add "inspections".

PAGE 27

4.5.1,f delete "Insulation resistance" and add: "Insulation resistance and dielectric tests".

PAGE 37

4.6.29 Delete entirety and add: "4.6.29 Insulation resistance and dielectric tests.

Add: 4.6.29.1 Insulation resistance. A potential of 500 VDC shall be applied for ten seconds between isolated pins of the connector and between pins of the connector and the indicator case. The resistance shall be not less than 10 megohms.

4.6.29.2 Dielectric. A potential of 200 VAC at a commercial frequency (60 Hz) shall be applied between isolated pins of the connector and between pins of the connector and the indicator case for a period of 60 seconds. Leakage current in excess of 0.05 ma, arc over, or breakdown shall constitute a failure."

4.6.30 Lines 2 and 3 delete "MIL-STD-810, Method 514" and add. "MIL-STD-810C Method 514.2". Change "figure 514.1-2" to read "figure 514.2-2".

Page 39

4.6.37 Line 2 delete "MIL-I-6181" and add: "MIL-STD-461 and MIL-STD-462".

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* 4.6.38 Line 2 and 3 change "MIL-STD-810, Method 501" to "MIL-STD-810C, Method 501.1".

* 4.6.39 Line 2 change "MIL-STD-810, Method 503" to "MIL-STD-810C, Method 503.1".

* 4.6.40 Line 3 change "MIL-STD-810, Method 504; Class 2 equipment" to "MIL-STD-810C, Method 504.1, equipment category 6".

PAGE 41

* 4.6.42 Lines 1 and 2 change "MIL-STD-810, Method 510" to "MIL-STD-810C, Method 510.0."

* 4.6.43 Line 2 change "MIL-STD-810, Method 507" to MIL-STD-810C, Method 507.1."

* 4.6.44 Line 2 change "MIL-STD-810, Method 508" to "MIL-STD-810C, Method 508.1".

* 4.6.45 Line 2 change "MIL-STD-810, Method 509" to "MIL-STD-810C, Method 509.1".

* 4.6.46 Line 2 change "MIL-STD-810, Method 506' to "MIL-STD-810C, Method 506.1".

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* 4.6.48 Lines 2 and 3 change "MIL-STD-810, Method 514, procedure I, part I, figure 514.1-2" to "MIL-STD-810C, Method 514.2, procedure I, part I, figure 514.2-2".

* 4.6.49 Line 2 change "MIL-STD-810, Method 516, procedure I, figure 516.1-2" to "MIL-STD-810C, Method 516.2, procedure I, figure 516.2-1, amplitude a, time duration c".

* 4.6.50 Change "MIL-STD-810" to MIL-STD-810C".

4.6.53 Delete entirety and add new: "4.6.53 Qualification reliability. The six test samples specified in 4.4.1 shall be subjected to a reliability test in accordance with MIL-STD-781C, test plan IC and vibration shall be 1.0g ± 10 percent. The test will run a minimum of 12,000 hours and will allow for no more than eight failures as shown in figure 14. This test demonstrates a lower test MTBF of 1000 hours at a 90 percent confidence level. Failure categories, analysis, repair and reporting shall also be in accordance with MIL-STD-781C."

PAGE 44

4.6.54, a and b delete entirety and add new: "a. The accept/reject criteria shall be in accordance with MIL-STD-781C, test plan IC.

b. If an accept decision cannot be made after accumulation of 12,000 total operating hours or if a reject decision is reached, testing shall be stopped. Further equipment deliveries shall not be made until corrective action has been taken and incorporated into all units. The effectiveness of the corrective action shall be demonstrated by testing as directed by the qualifying activity."

PAGE 45

4.6.56 Delete entirety and add new: "4.6.56 Maintainability demonstration. Unless otherwise specified (see 6.2), the indicator shall be subjected to a

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maintainability demonstration in accordance with MIL-STD-471A, test method 8. A maintainability test plan in accordance with Section 4 of MIL-STD-471A shall be submitted and approved by the qualifying activity prior to initiating the maintainability demonstration test. The mean maintenance time (Mean) shall not exceed 240 minutes not including the time to seal and unseal the unit. The maximum time (Mmax) shall not exceed 360 minutes not including the time to seal and unseal the unit. The maximum ratio of percentile to mean is 95 percent which corresponds to a producer's risk α (Alpha) and consumer's risk β (Beta) of .05."

4.7 Delete entirety and add new: "4.7. Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification."

* Delete Section 5 in its entirety and add:

"5. Packaging

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial The Industrial preservation of the indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

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5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129."

* 6.2 Delete "title and first sentence" and add: "Acquisition requirements. Acquisition documents must specify the following:".

* 6.2.c Delete entirely.

* Add "6.2,h Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents refrenced (see 2.1)".

PAGE 47

6.4: Delete last sentence and add: "The activity responsible for the Qualified Products list is OC-ALC/MMEDO, Tinker AFB OK 73145-5990, and information pertaining to qualificaton of products may be obtained from that activity."

* Delete "6.5" entirely and add new "6.5". International standardization agreement. Certain provisions of this specification are the subject of international standardization agreements NAT-STD-3637 and NAT-STD-3640, and ASCC Air Standard 10/42 and 10/54. When amendment, revision or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accomodations."

* Delete "6.6" entirely.

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* "Delete figure 14 and add new figure 14 attached to this amendment".

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MIL-I-27619G
AMENDMENT 3

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment are made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
AIR FORCE - 99
NAVY -AS

Preparing Activity:
AIR FORCE -71

Project No:
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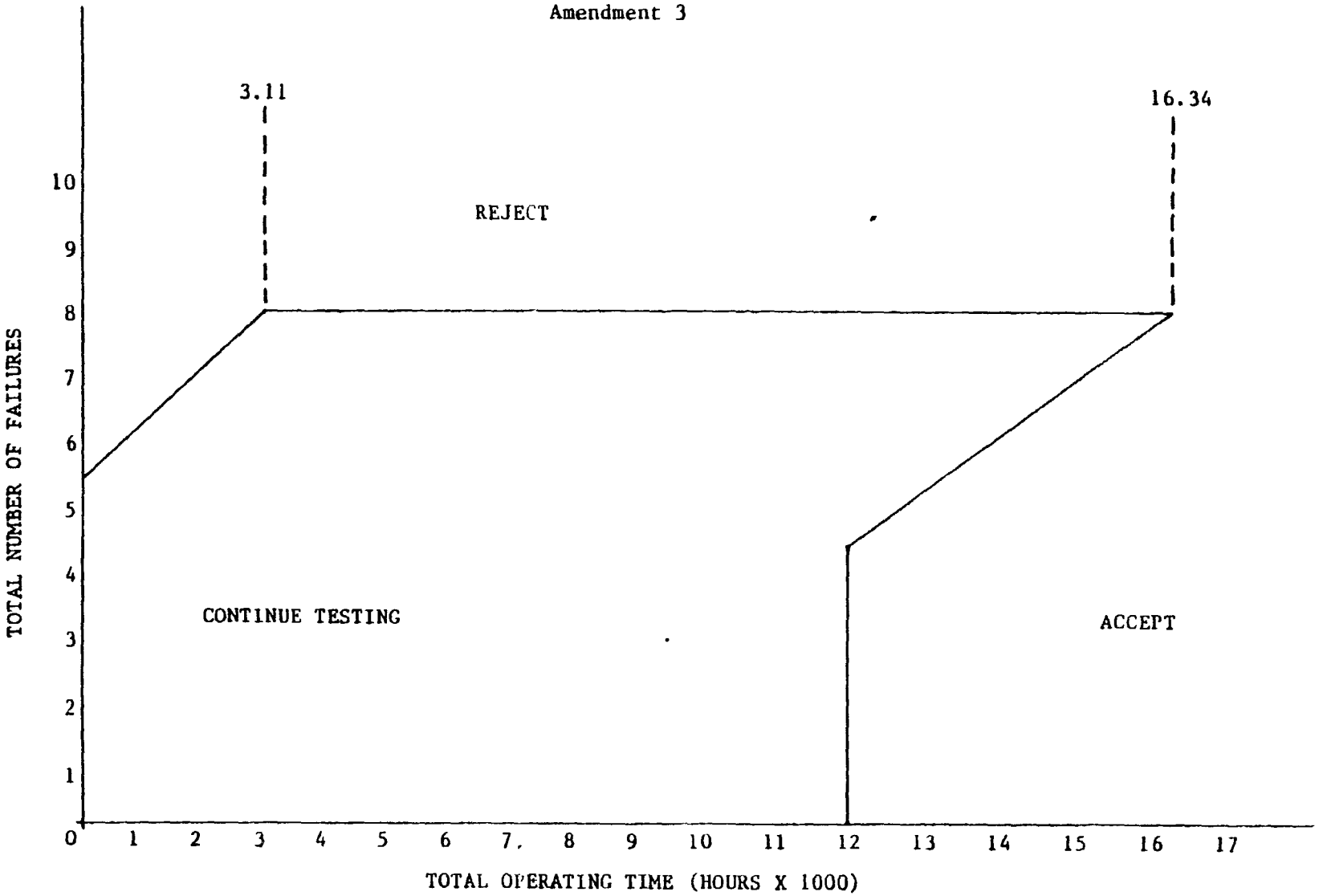


Figure 14. Qualification/reliability accept/reject criteria.

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